REFERENCES CITED

ALDEN, W. C., and STEBINGER, E., 1913, Pre-Wisconsin glacial drift in the region of Glacier National Park, Montana: Geol. Soc. America Bull., v. 24, p. 529–572.

ANTONIUK, S. A., 1954, A mineralogic study of the Saskatchewan sands and gravels: Unpublished M.Sc. thesis, University of Alberta.

CALHOUN, R. H. H., 1906, Montana lobe of the Keewatin ice sheet: U.S. Geol. Survey Prof. Paper 50.

COLEMAN, A. P., 1909, The drift of Alberta and the relations of the Cordilleran and Keewatin ice sheets: Royal Soc. Canada Trans., v. 3, ser. 3, sec. 4, p. 3–12.

DAWSON, G. M., 1895, Note on the glacial deposits of south-western Alberta: Jour. Geology, v. 3, p. 507–511.

DENNY, C. S., 1936, Periglacial phenomena in southern Connecticut: Am. Jour. Sci., v. 32, p. 322–342.

DREIMANIS, A., 1960, Pre-classical Wisconsin in the eastern portion of the Great Lakes region, North America: Rept. Internat. Geol. Congress, 21st sess., Norden, pt. 4, p. 108–119.

GRAVENOR, C. P., and BAYROCK, L. A., 1961, Glacial deposits of Alberta. *In* Soils in Canada: Royal Soc. Canada Special Pub. no. 3, p. 33–50.

——— and ELLWOOD, R. B., 1956, A radiocarbon date from Smoky Lake, Alberta: Research Council of Alberta, Prelim. Rept. 56-3.

HORBERG, L., 1952, Pleistocene drift sheets in the Lethbridge region, Alberta, Canada: Jour. Geology, v. 60, p. 303–330.

LACHENBRUCH, A. H., 1960, Thermal contraction cracks and ice wedges in permafrost: U.S. Geol. Survey Prof. Paper 400-B, p. 404–406.

LEFFINGWELL, E. DE K., 1919, The Canning River region, northern Alaska: U.S. Geol. Survey Prof. Paper 109.

MCCALLUM, K. J., and DYCK, W., 1960, University of Saskatchewan radiocarbon dates II: Am. Jour. Sci., Radiocarbon Supplement, v. 2, p. 73–81.

MCCONNELL, R. G., 1885, Report on the Cypress Hills, Wood Mountain and adjacent areas: Geol. Survey Canada Ann. Rept., v. 1, pt. C, p. 1–78.

POPOV, A. I., 1955, The origin and development of massive fossil ice: Materialy kosnovam ucheniya o merzlykh zonakh zemnoi kory. Vypusk 2. Akademiya Nauk SSSR, Institut Merzlotovedeniya imeni V. A. Obrucheva. English translation in Natl. Research Council of Canada, Tech. Translation 1006, p. 5–24, 1962.

RUSSELL, L. S., and LANDES, R. W., 1940, Geology of the southern Alberta plains: Geol. Survey Canada Mem. 221.

RUTHERFORD, R. L., 1937, Saskatchewan gravels and sands in central Alberta: Royal Soc. Canada Trans., v. 31, ser. 3, sec. 4, p. 81–95.

SCHAFER, J. P., 1949, Some periglacial features in central Montana: Jour. Geology, v. 57, p. 154–174.

SHARP, R. P., 1942, Periglacial involutions in northeastern Illinois: Jour. Geology, v. 50, p. 113–133.

TABER, S., 1943, Perennially frozen ground in Alaska: Its origin and history: Geol. Soc. America Bull., v. 54, p. 1433–1548.

TYRRELL, J. B., 1886, Report on a part of northern Alberta: Geol. Survey Canada Ann. Rept., v. 2, pt. E, p. 139.

ROCK MAGNETISM AND DIKE CLASSIFICATION[1]

DAVID W. STRANGWAY
University of Colorado, Boulder, Colorado

ABSTRACT

It is shown in this paper that it is possible to use remanent magnetism as a tool for geologic classification, Several suites of diabase dikes from Ontario and Quebec were sampled and widely differing magnetic properties found. It is shown that the direction of remanent magnetization, as well as the intensity, forms separate and distinct criteria for distinguishing between swarms. An earlier paper indicated that the direction of stable remanent magnetization is distorted by the shape of the dike. In spite of this distortion, widely differing directions of remanent magnetization indicate that the magnetic field of the earth existed in Precambrian times and that it wandered considerably in that era.

INTRODUCTION

There are many problems in geology but only a few satisfactory tools to help solve these problems. In recent years the study of remanent magnetism has been making significant contributions. It is now well established that the magnetic pole of the earth has wandered in relation to the continents, and a great deal of data extending back to the end of Precambrian times has established a polar-wandering curve for North America and Europe. In this study several

[1] Manuscript received January 9, 1962.

This content downloaded from 129.100.058.076 on October 26, 2018 00:31:30 AM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

Precambrian dikes have been sampled, and it will be shown that these data can be used to correlate the dikes. In the Canadian Shield there are several well-established swarms of dikes which are described in the literature as younger Precambrian dikes. Unfortunately there have been few attempts to determine the age of the dikes. In the present paper it will be shown that rock magnetism is a simple and effective tool allowing unique identification of the dikes in various swarms.

In addition, it will be shown that apparent polar-wandering extends far back into middle Precambrian times, and therefore the liquid core of the earth must have existed since that time at least.

CLASSIFICATION OF DIKES

Dike swarms seem to be of two major types—one of these is a swarm of parallel dikes apparently unrelated to any known center of volcanic activity. Other swarms seem to occur radiating outward from some known center of igneous activity. None of the latter seems to occur in the area discussed in this paper although some workers have classified the Sudbury dike swarm as being of this type.

Richey (1939) studied the dikes of Scotland in detail. In this study an excellent description of several dike swarms is presented. In Scotland the relative ages of the dikes could be determined with some accuracy by the strata they intruded. In the Canadian Shield area, with one exception, there are no overlying strata to permit classification, so it is necessary to resort to other, less obvious criteria for classification.

The most useful of these is the strike of the dike in relation to that of other dikes, but this is not always adequate, as swarms of different ages may have the same strikes. Anderson (1951) has made a thorough study of the dikes of Scotland and the mechanics of their intrusion. He points out that dikes tend to have a fairly constant width and to be equally spaced. This means that width and separation may also be useful criteria for dike classification.

One other method of classification has been a careful study of composition, a criterion useful in only a few particular cases. The usual classification is one where we have olivine diabase, quartz diabase, or simple diabase. In the process of crystallization of diabasic magma it has been found that olivine and quartz are mutually exclusive, and it is supposed that the dikes of various compositions are formed from different magmas. Unfortunately quartz and olivine are only present in small quantities in dikes, and careful thin-section study is required to establish whether these minerals are present. In compiling information on the younger Precambrian dikes of Canada it is clear that composition studies have not often been made.

It can safely be said that where there is no sedimentary sequence of rocks, it is not at the present time possible to classify dikes with any degree of certainty. The study of magnetic properties gives us another tool permitting a definite classification.

DIKE SWARMS OF NORTHWEST QUEBEC AND NORTHEAST ONTARIO

In the Precambrian Shield of Quebec and Ontario there are several well-developed dike swarms. In parts of the area, in particular the commercially significant greenstone belts, detailed geologic mapping has been done. Recently, wide-scale aeromagnetic surveys have been carried out, showing the presence of a large number of dikes. In the present study a careful compilation of all reported dike exposures and all published airborne magnetic data of a portion of Ontario and Quebec has been made. In the maps accompanying this paper (figs. 1, 2) the compiled data are presented. It can be seen quite clearly that there are several different dike swarms with different strike directions. The dikes are discussed by Norman (1948), by Moore (1955), by Gill and L'Esperance (1952) and in the reports of the Ontario Department of Mines, the Quebec Bureau of Mines, and the Geological Survey of Canada. In most references the diabase dikes are described as being the youngest rock present. Occasionally they are classified

This content downloaded from 129.100.058.076 on October 26, 2018 00:31:30 AM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

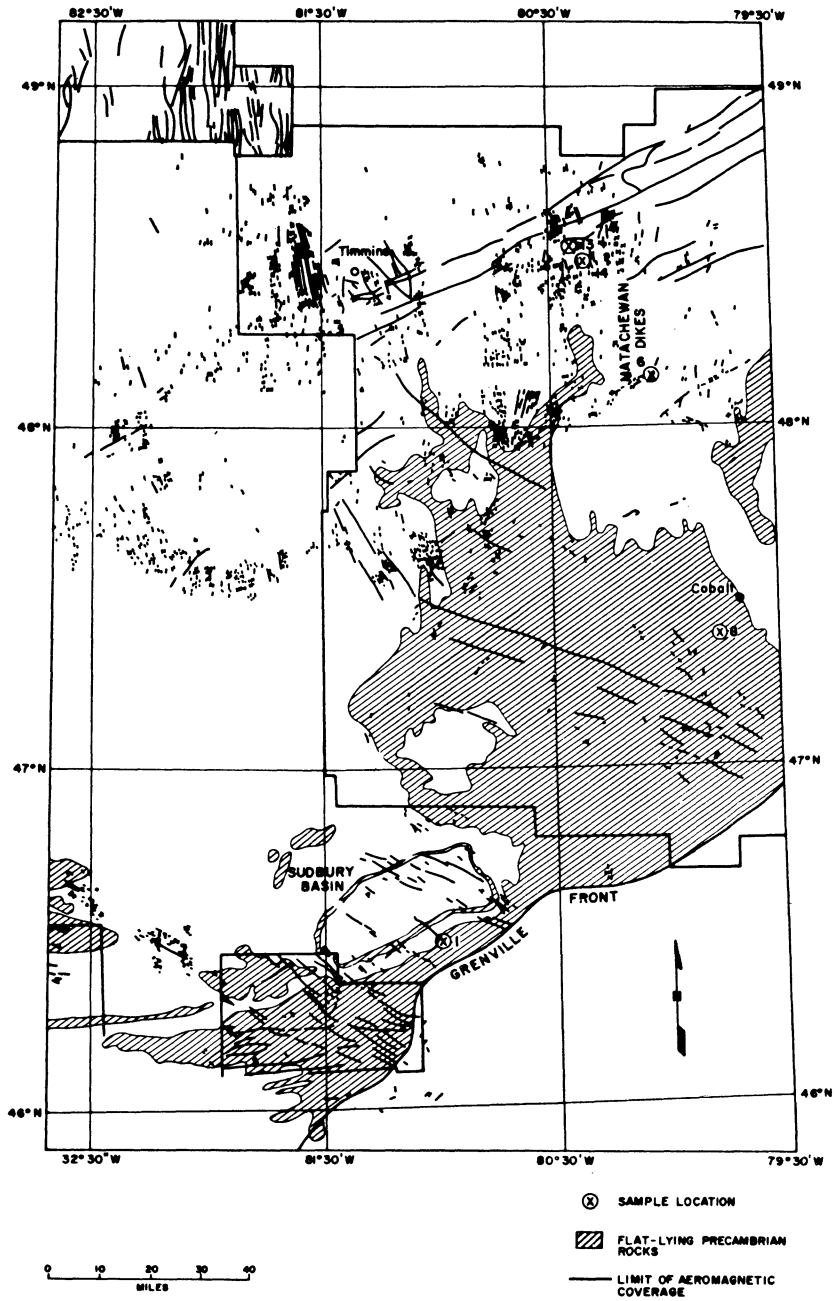


FIG. 1.—Diabase dikes of northwestern Ontario

This content downloaded from 129.100.058.076 on October 26, 2018 00:31:30 AM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

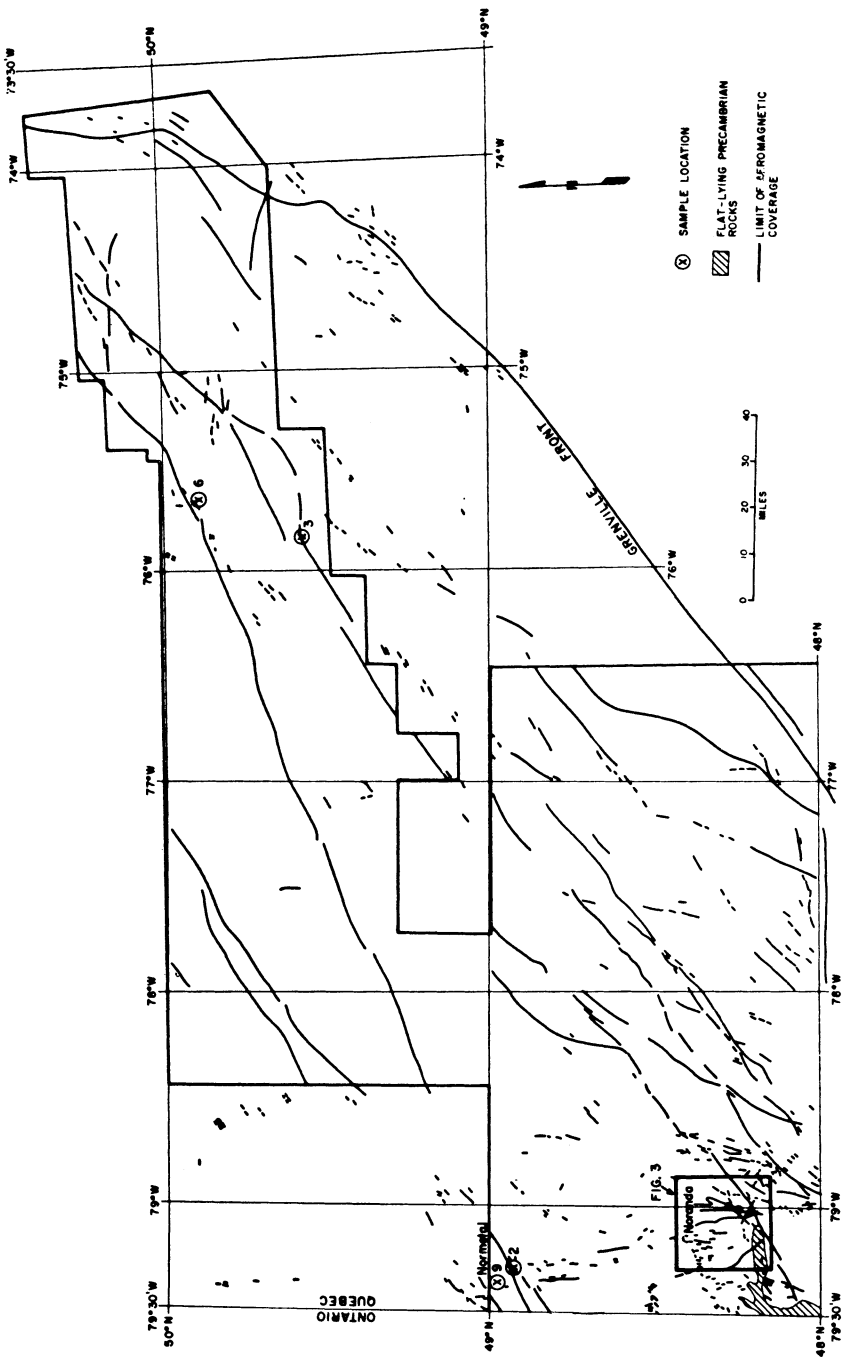


FIG. 2.—Diabase dikes of northwestern Quebec

This content downloaded from 129.100.058.076 on October 26, 2018 00:31:30 AM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

as containing olivine or quartz. Unfortunately this observation has often been based on inference with nearby dikes so that it is commonly not possible to determine the composition of these dikes from published reports. In several cases two exposures were clearly defined by aeromagnetic surveys as being part of the same dike, and yet published reports classified the two exposures as being of olivine and of quartz composition, respectively. It seems probable, however, that any one dike is of uniform composition. If the composition along strike is not uniform, then, of course, composition is not a useful criterion.

Study of the literature reveals the presence of the following probable swarms of dikes. The classification in several cases is not clear, and it will be shown that the magnetic properties can be very useful in helping to clarify the interrelationships of these dikes.

MATACHEWAN DIKES

One of the most clearly defined sets of dikes is the Matachewan swarm. As shown in figure 1, the Matachewan dikes are a swarm with a north-south strike. These are probably the oldest dikes, as they can be seen to be cut off by the younger flat-lying Precambrian rocks exposed in the portions of the shield near the Grenville Front. These younger flat-lying rocks are the Cobalt sediments and the Nipissing diabase sill. The age of these flat-lying rocks is not yet fully agreed upon, but they are generally considered to be about 1.5 billion years. Where the dikes have been carefully mapped, as for example in the Timmins area (Hogg, 1954), it is clear that there are many hundreds of them present. Detailed ground magnetometer surveys outline these dikes very clearly.

Careful study of these dikes and ground magnetic surveys reveal that the dikes have been subjected to a considerable amount of faulting so that several cycles of tectonic activity have taken place since the dikes were first emplaced. From a study of the mechanics of dike intrusion it is seen that in the process of dike intrusion a dike can be deflected along a plane of weakness. When this is the case, however, the dike is continuous through this plane. In the case of the Matachewan dikes it is often not possible to trace the dike through fault zones. In other words, it is clear that much of the east-northeast faulting found in northern Ontario and Quebec is post-Matachewan in age.

The composition of these dikes is that of normal quartz diabase. Some writers have attempted to classify them on the basis of a porphyritic nature. It does not seem from the present compilation that this is a useful criterion for classification. In figure 1, one area can be seen in which detailed aeromagnetic surveys revealed a large number of these dikes. Other nearby surveys unfortunately do not show them because flight lines were wide-spaced and parallel to the dikes. A recent study dates these dikes at approximately 2.5 billion years (Wanless and Fahrig, personal communication and IUGG meeting, Berkeley, 1963).

EAST-WEST DIKES OF NORTHERN MICHIGAN

A swarm of east-west–striking dikes has been mapped in northern Michigan. These dikes have not been described in great detail, but early aeromagnetic surveys revealed the presence of several of these negatively polarized dikes. Balsley, James, and Weir (1949) refer to these dikes, and Graham (1953) sampled them and measured their remanent magnetism in some detail. These dikes strike east-west parallel to the ancient Penokean mountain chain. James (personal communication) indicates that ground magnetometer studies show conclusively that these dikes are overlain by the Jacobsville Sandstone. Most workers agree that this Jacobsville Sandstone is late Precambrian in age (Keweenawan). The dikes cut Animikie rocks.

It is interesting to note that in the area north of Ottawa, Ontario, a series of east-west–striking dikes has also been reported. Although detailed airborne magnetic surveys have been carried out in the vicinity of Ottawa and flight lines were mainly parallel to the dike swarms, the dikes are not

This content downloaded from 129.100.058.076 on October 26, 2018 00:31:30 AM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

clearly outlined by these surveys and one can only speculate on their nature. Dating by Fahrig *et al.* dates these dikes at 790 million years.

LATER NORTH-SOUTH DIKES

In the vicinity of the Ontario-Quebec border there seem to be a few additional north-south dikes. It is not clear whether these dikes are related to the north-south Matachewan dikes, and so far no method of distinguishing between the two swarms has been reported. In particular, several dikes are known in the Noranda area which intersect the Noranda orebody or pass close to it (fig. 3). A north-south dike is also known in the Normetal area which cuts the Normetal orebody. As will be seen shortly in the discussion of results, the rock-magnetism data clearly distinguish this Normetal or Abana dike as being different from the Matachewan dikes. The other north-south dikes which were sampled in the Noranda area will be shown to be of Matachewan age. In the vicinity of Cobalt, Ontario, a few north-south–striking dikes have been reported cutting all the late flat-lying Precambrian rocks (Nipissing diabase and Cobalt).

DIKES PARALLEL TO THE GRENVILLE FRONT (E.NE.)

The Grenville Front has been a controversial geologic feature for many years, but it is now well established in the literature. It appears to represent a division between highly metamorphosed and less metamorphosed rocks. The Grenville subprovince lies on the southeastern edge of the Shield itself (see figs. 1, 2) and seems to represent a region in which mountain-building took place about one billion years ago. The time of formation of a set of east-northeast dikes is unknown, but it is clear from figures 1 and 2 that this swarm is parallel to the Grenville Front and is probably related to the mountain-building of Grenville times. Fahrig *et al.* (personal communication) have dated these large dikes at 1.23 billion years.

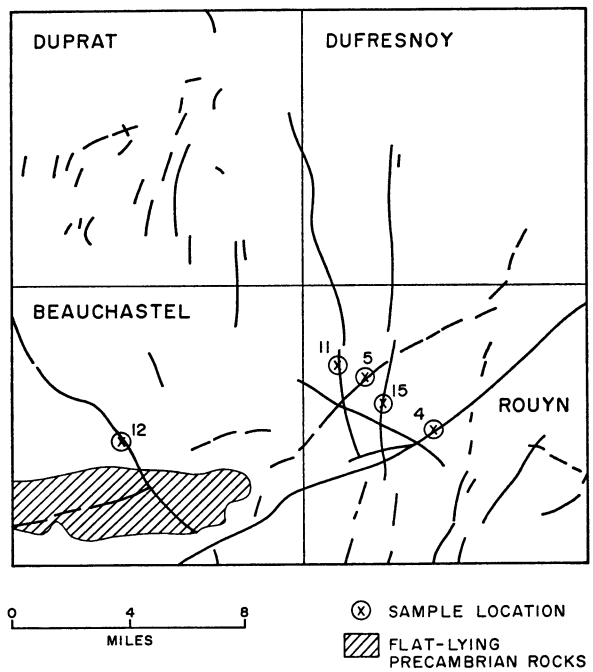


FIG. 3.—Diabase dikes of the Noranda area

This content downloaded from 129.100.058.076 on October 26, 2018 00:31:30 AM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

There are not a great number of these dikes present, but those that are found are often several hundred miles long (as shown by aeromagnetics) and are usually 300–500 feet wide. It seems clear that they were formed from a liquid magma which had a large areal distribution. It is inconceivable that the material could have flowed for several hundred miles from some center of activity without becoming chilled by the host rock. This leads one to think that the dikes all formed from a single large sheet distribution of molten magma and to think that the dikes should therefore all be of the same composition—either olivine-rich or quartz-rich. This is definitely not found to be the case. In the vicinity of Noranda, Quebec, for example, two parallel dikes are mapped, one being of quartz and one of olivine composition. It is not possible to prove, but, from the evidence of rock magnetism, it appears that these two dikes are of roughly the same age.

The only place where it appears to be possible to get a relative age for dikes of this type is a few miles southeast of Noranda, Quebec, where one of the dikes evidently stops at the Cobalt Front (limit of flat-lying Precambrian rocks). At first glance it would seem that the Cobalt sediments postdate the dike, but closer study of the aeromagnetic maps shows that the dike, although continuous for many miles, fails to appear about five miles to the east where it would be expected. It is not clear then whether the dike truly stops at the Cobalt Front or whether it is continuous beneath the Cobalt sediments (fig. 2).

There has been some speculation that the Nipissing diabase sill found in the vicinity of Cobalt, the norite of the Sudbury Basin, and these east-northeast dikes have a common origin. The conclusion is based on compositional studies. If this is the case, it then becomes clear that the dikes are post-Cobalt, as the Nipissing diabase sill intrudes the Cobalt sediments.

NORTHWEST DIKES OF SUDBURY AREA

In the vicinity of the Sudbury Basin a swarm of olivine dikes has been mapped; and, where aeromagnetic coverage is available, a large number of additional dikes can be seen to the south of the basin. Similar dikes are also found to the northeast, where dikes with the same strike are found to cut the flat-lying Precambrian rocks (fig. 2). It is evident that these dikes are the youngest rocks in the area. One age determination has been reported by Fairbairn, Hurley, and Pinson (1960) who quote an age of 1.02 billion years for one of these dikes using the potassium-argon method. Later work by Fahrig *et al.* has dated these dikes at 1.22 billion years.

OTHERS

West of the Sudbury Basin in the vicinity of Elliot Lake, Moore and Armstrong (1943) have reported a swarm of dikes with a west-northwest strike. These dikes are of simple diabase and are about 50 feet wide. In the present study no samples of these dikes were taken, and so they are not discussed further. A few other dikes have been reported but none that could be considered to be a swarm are discussed in the literature.

DIRECTION OF REMANENT MAGNETISM

A total of one hundred eighty-six samples each giving several cubes were collected from fourteen dikes representing several of these dike swarms. The results are tabulated in tables 1 and 2, where data about intensity and direction of magnetization are tabulated. The remanent magnetism (Jr) was measured and the directions plotted. A considerable scatter to the results was found, but, after demagnetizing in a.c. fields of 100–200 oersteds, the results from any one dike were generally found to group fairly well. In a previous paper (Strangway, 1961) it was observed that this stable magnetization was usually in the plane of the dike. In this paper, however, variations in direction from dike to dike are discussed and found to have significance. In table 1 the declination and inclination of stable remanent magnetism of each of the sampled dikes are tabulated. In this table it can be seen that the various swarms show different directions of magnetization. Dikes 2–6, for example, all strike

This content downloaded from 129.100.058.076 on October 26, 2018 00:31:30 AM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

east-northeast, and we see that each of these has a declination between 246° and 265° and an inclination between +42° and +61°. Each of these dikes evidently belongs to the same swarm, and, even though some of these are olivine diabase and some quartz diabase, there is no reason to distinguish between them on the basis of their direction of magnetization. The column marked "α" represents the half-angle subtended by the circle of confidence at a 95 per cent level using Fisher statistics. The column marked K is a precision parameter. A mean inclination of 51° falls within the confidence limit of each

TABLE 1

DIRECTION DATA

No.	Dike	Strike	Composition	No. of Samples	α	K	D	I*	Comments
1...	Sudbury	297	olivine	9	25.0	5.1	265	+ 8R†	No stable magnetization found
2...	Normetal	251	quartz	13	8.8	24.5	265	+42	Probably all of same age
3...	Opawica	258	olivine	9	10.8	23.5	263	+56	
4...	Noranda	256	olivine	7	15.9	12.8	256	+61	
5...	Noranda	226	quartz	5	18.0	12.9	246	+47R†	
6...	L. la Treve	E.NE.	olivine	1	.....	.....	233	+69	
7...	Sudbury (norite)	sill	.....	large no. (Hood)	‡	.....	245	+82	
							300	+78	
8...	Nipissing	sill	quartz	1	.....	.....	180	+65	
9...	Abana	16	simple	31	5.3	24.2	18	+30R†	
10...	Michigan	85	?	large no. (Graham)	.....	.....	82	−86	
11...	L. Dufault	164	quartz§	8	16.1	12.5	161	−19	Probably all of same age
12...	Aldermac	154	quartz§	5	40.5	1.2	190	−10	
13...	Matachewan	176	quartz§	21	19.2	6.5	186	− 3R†	
14...	(A+B)	197	.....	.....	.....	.....	.....	.....	
15...	Noranda	191	simple?	8	25.9	4.9	215	− 7	
16...	Teck-Hughes	N.-S.	quartz	1	.....	.....	216	− 5	

* In a previous paper (Strangway, 1961) the sign of the inclinations was inadvertently reversed.
† At least one reversed sample found.
‡ (*a*) N.+S. equal tilting; (*b*) N., 10° tilt., S., 30° tilt.
§ Micropegmatite present.

TABLE 2

INTENSITY DATA

Dike No. (by Increasing Age)	Strike	Dike Name	No. of Samples	Average Qr	Jt/Jr	$(Qn)/Jt/Jr$	$(Qn)/Jt/Jr$ (Normalized)
1.....	NW.	Sudbury	39	0.40	6.6	.05	.040
2.....	E.NE.	Normetal	58	4.0	5.2	.77	.24
3.....	E.NE.	Opawica	61	4.2	6.3	.67	.20
4.....	E.NE.	Noranda	52	2.3	3.4	.67	.20
5.....	E.-W.	Noranda	37	0.58	.....	.....	.....
9.....	N.-S.	Abana	60	1.0	4.6	.21	.11
10.....	E.-W.	Michigan	.....	10.0	10.0	1.0	.25
11.....	N.-S.	L. Dufault	32	0.14	7.3	.019	.010
12.....	NW.	Aldermac	21	0.25	11.2	.023	.019
13, 14.....	N.-S.	Matachewan (A+B)	78	0.40	23.0	.014	.014
15.....	N.-S.	Noranda	29	0.45	28.8	.015	.013
Present-day (average of all heated samples)			20	6.0	1.0	6.0	1.5

This content downloaded from 129.100.058.076 on October 26, 2018 00:31:30 AM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

of these dikes except Dike No. 6, represented by a single sample.

No stable magnetization could be found in the case of the Sudbury olivine dike. One possible explanation is that the dike was formed at a time when the earth's field was close to normal to the plane of the dike (Strangway, 1961). It is interesting to note in this connection that the Sudbury dikes and the east-northeast ones (Nos. 1–6 in Tables I and II) are given the same age. This means that if the horizontal component of the magnetic field was parallel to one swarm it was perpendicular to the other.

Dike No. 9 is a north-south dike that could readily have been confused with other north-south dikes. This particular dike was sampled both at surface and underground where it intersects the Normetal orebody. It has a mean declination of $+18°$ and a mean inclination of $+30°$. For Dike 10 a set of data collected by Graham (1953) from the east-west dikes of northern Michigan is presented. In this case we see a remanent magnetization with a declination of 82° and an inclination of $-86°$. The remaining dikes, 11–16 inclusive, were of unknown relationship except that 13 and 14 were collected from an area near the town of Matachewan, the classic area for the Matachewan diabase dikes. These dikes are all north-south in strike, and from the remanent magnetism directions there seems to be little question that they are all from the same swarm. An average inclination of $-10°$ falls within the circle of confidence of all the Matachewan north-south dikes. The use of the direction of remanent magnetism has thus enabled us to distinguish very definitely between the various dike swarms.

In figure 4 a plot of declination versus inclination is presented. In this figure the results of Precambrian remanent-magnetism studies from the Lake Superior-Ontario-Quebec area are presented. It can readily be seen that a fairly smooth curve can be drawn through all the points. The flat-lying Keweenawan rocks in the vicinity of Lake Superior were reported by DuBois (1962), and data from the Sudbury Basin and basic intrusives in the Grenville subprovince were reported by Hood (1958).

These geologic units are in their correct time sequence. It is interesting to observe that the Duluth gabbro and some intrusives in the Grenville province (Boulter and Tudor gabbro—Hood, 1958) show similar directions of magnetization indicating an approximate Keweenawan age for all these intrusives. Moving back along the curve, we find that the east-northeast dikes tend to fall in a group. This indicates that these dikes are probably pre-Keeweenawan in age although they have generally been classified as Keweenawan. It also indicates that these east-northeast dikes probably predate the Grenville orogeny by a short interval of time. The Grenville is commonly dated at 1,000 million years (Lowdon, 1961) or about the same as the Duluth gabbro dated at 1,120 million years (Goldich, Nier, Baadsgaard, Hoffman, and Krueger, 1961).

The curve becomes rather indefinite at this point due to lack of data; but the Sudbury (*a*) model of Hood shows that the direction of remanent magnetization in the Sudbury Basin and the east-northeast dikes is about the same, indicating a probable similar age for the two units. A single sample from the Nipissing sill gives a direction which indicates a slightly older age. Detailed sampling of the Nipissing sill should help clarify the relationship with the Sudbury Basin and the east-northeast dikes.

In other parts of figure 4 the Abana north-south, Michigan east-west, and the Matachewan north-south dikes are found.

The exact directions of remanent magnetization were shown in an earlier paper (Strangway, 1961) to be distorted by the anisotropy of the dike. This means that the direction of magnetization recorded probably represents the projection of the field into the plane of the dike rather than the true direction of the ancient earth's field. In any event, the remanent magnetism has clearly shown that the various swarms can readily be classified in this manner, as each group of dikes has a distinct direction of remanent magnetization.

This content downloaded from 129.100.058.076 on October 26, 2018 00:31:30 AM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

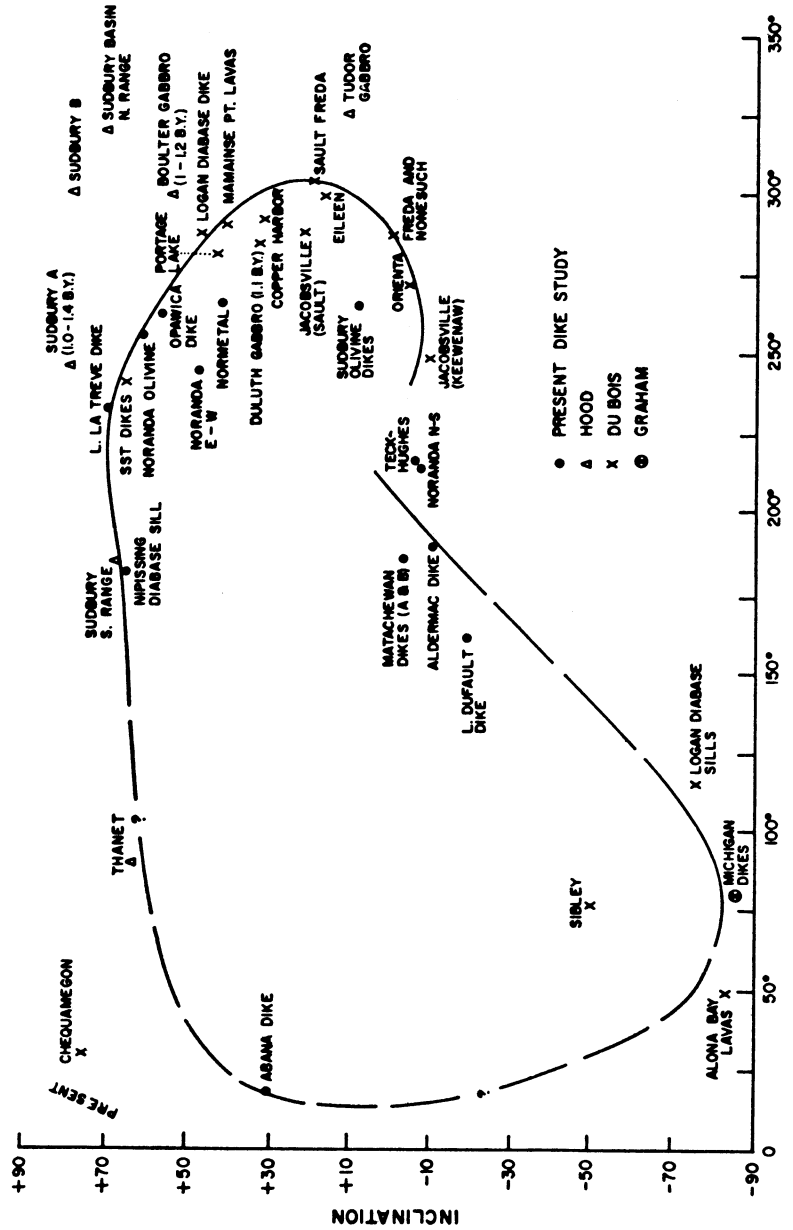


FIG. 4.—Direction of permanent magnetism—Precambrian data

This content downloaded from 129.100.058.076 on October 26, 2018 00:31:30 AM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

INTENSITY OF MAGNETIZATION

A study of the intensity of magnetization has also been made. In table 2 the data on intensity have been tabulated. The intensity of remanent magnetism (Jr) and the susceptibility (X) have been used to calculate the Koenigsberger ratio, $Qn = Jr/XH$ where H is the present earth's field. The dikes are numbered as before. Data from a total of 430 cubes have been tabulated. Except for the Sudbury olivine dike we see that the Qn ratio tends to decrease as we move down the table. Dikes 2–5, all belonging to the east-northeast swarm, show values of Qn between 2.0 and 4.2, while the older Matachewan dikes show values of Qn between 0.14 and 0.45. (It will be remembered that dikes 11, 12, 15 were not previously known to be of Matachewan age.) The value Qn for the Michigan east-west dike has been estimated by Van Voorhis (1964) from a study of the magnetic properties and magnetometer profiles over the dikes. These studies give a value of Qn equal to 10.0. In order to emphasize the variation in Qn with dike swarm the chart shown in figure 5 has been plotted. We see clearly that the east-northeast dikes have large values of Jr, while the older Matachewan dikes show smaller values of Jr on the average.

In addition to the use of the quantity Qn, we may also use the ratio of the remanent magnetism acquired by cooling from above the Curie temperature in the present earth's field (Jt) to that of the remanent magnetism found in the fresh sample. The ratio Jt/Jr tabulated in table 2 shows a general tendency to increase with increasing age. Ignoring the Sudbury dike again, we see that Jt/Jr for the east-northeast dikes is between 3.4 and 6.6, while that of the Matachewan dikes is between 7.3 and 28.8. In column (3) the ratio $Qn/Jt/Jr$ has been tabulated. It is seen that the various swarms show a large variation in this quantity. In table 2 the values Qn and Jt/Jr, which these dikes would have if they had cooled through the Curie point under present-day conditions, are also shown. The ratio $Qn/Jt/Jr$ has a value of 5.9, much greater than any of the dikes studied.

The Sudbury north-west dike shows results which are inconsistent with those of the other dikes. It is probable that this dike has a distinctly different mineralogy from the others. Archbold (personal communication) reports that the Sudbury dikes contain appreciably more magnetite than the other dikes, and so the magnetic relaxation process may be different in the case of these dikes.

It is known that the intensity of magnetization acquired by a specimen is proportional to the field at the time of cooling (Neel, 1955). The value of $(Jt/Jr)/Qn$ needs to be normalized to some standard latitude so that the intensity data may be compared. The field strength depends upon latitude and the value of $Qn/(Jt/Jr)$ is shown in table 2 normalized to an equatorial latitude using the inclination of the magnetization shown in table 1. Assuming the Matachewan dike has an age of 2.5 billion years and the E.NE. dikes an age of 1.23 billion years, a graph of $(\log t)^2$ versus $Qn/(Jt/Jr)$ has been plotted in figure 6. The plot versus $(\log t)^2$ is used since it is expected that the remanent magnetization should depend on the logarithm of time (Neel, 1955). Data from the other dikes is then added to the graph using the normalized value of $Qn/(Jt/Jr)$ and if it is assumed that the magnetic relaxation process is similar for the various dikes, the age can be determined.

This procedure sets the age of the Abana dike at 1.95 billion years and the age of the Michigan east-west dikes at approximately 1 billion years depending upon the exact value of Jt/Jr used. These values are quite consistent with other geologic facts. Using an extrapolation of the curve to the $Qn/(Jt/Jr)$ value measured for the present-day samples an age of 10 million years is estimated. Compared with the time scale used in this study this age is very close to the present and so the procedure seems to be a useful one.

There seems to be no doubt that the intensity of magnetization of rocks of similar

This content downloaded from 129.100.058.076 on October 26, 2018 00:31:30 AM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

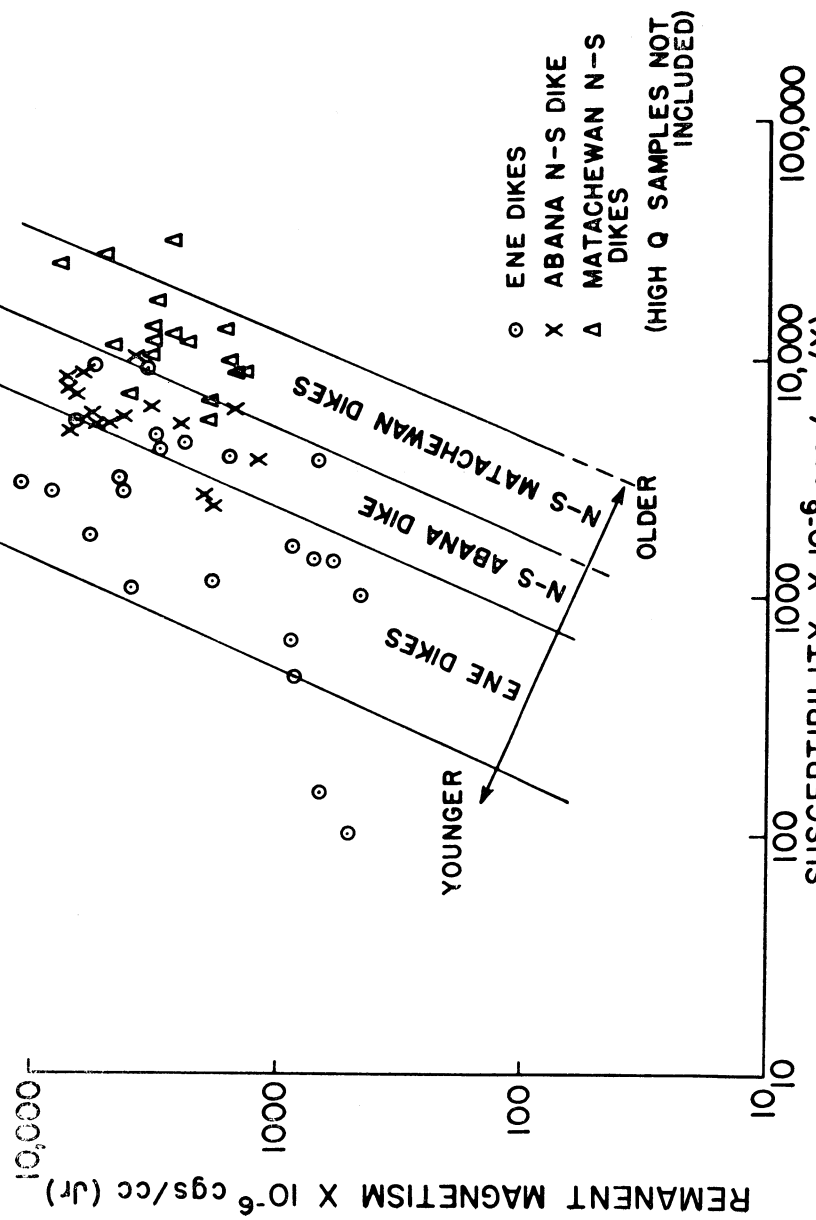


FIG. 5.—Remanent magnetism versus susceptibility

This content downloaded from 129.100.058.076 on October 26, 2018 00:31:30 AM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

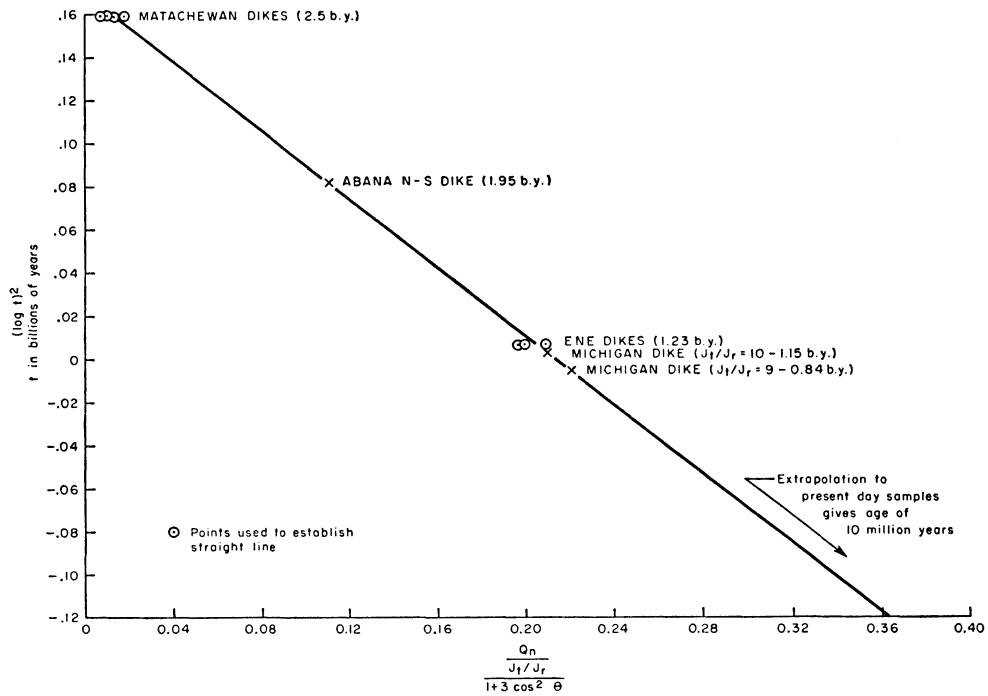
composition can be used as a tool for classifying rocks and assigning an older or younger label to rocks of similar composition but widely different age. Thus paleomagnetism gives us two distinct criteria—direction and intensity—for classifying these dikes.

PALEOMAGNETIC IMPLICATIONS

The classification of these dikes using the intensity and direction of remanent magnetism has a clear dependence on the magnetic field of the earth. As discussed in an earlier paper (Strangway, 1961), there is an indication that the shape of the dike has led to a distortion of the stable component of remanent magnetism preserved in these dikes. Thus the true earth's magnetic field direction may not be preserved in the dikes. However, it was also shown in the earlier paper that, if the earth's magnetic field does not lie close to the plane of the dike, the magnetic field intensity is such that no stable component of remanent magnetism is formed. In the Sudbury dike reported in this paper, for example, no stable magnetization could be found.

In figure 7 and in table 3, Precambrian virtual magnetic pole positions determined from North American rocks are plotted. These data have been taken from Cox and Doell (1961) who have given an excellent compilation of paleomagnetic data. Recent data from Missouri have also been included (Hays, Scharon, and Anderson, 1962). Pole positions have been calculated for the dikes sampled in this study with the realization that anisotropy may cause a displacement in the apparent pole position. It is interesting to observe that the magnetic pole appears to have made a loop in the Pacific during late Precambrian times. This is similar to a loop reported recently by Creer (1962) in a study of South American data.

The east-northeast dikes (2, 3, 4, 5) give



RELATION BETWEEN INTENSITY DATA AND AGE

FIG. 6.—$(\log t)^2$ vs. $(Jt/Jr)/Qr$

This content downloaded from 129.100.058.076 on October 26, 2018 00:31:30 AM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

pole positions close to those measured by DuBois for the Keweenawan or late Precambrian rocks. We are led to the conclusion that these dikes give pole positions which have a paleomagnetic significance and that the dikes predate the late Precambrian rocks of Lake Superior. We might also point out that, since the magnetization lies in the plane of the dike and that the direction may be a true one, the dike swarm may have been formed in a north-south direction. The data also correlate well with the measurements of Hood (1958) on the Sudbury Basin. This leads to the observation that the east-northeast dikes and the Sudbury Basin may have a common age.

Because of the wide variation in pole positions there seems to be little doubt that the earth's field moved with respect to North America not only in late Precambrian times, but as far back in time as 2.5 billion years ago. It has not been possible to draw any conclusions about the intensity of the earth's magnetic field in the past except for the studies by Thellier and Thellier (1959) where some inferences about the very recent field have been made. The data presented in this study show that the sampling extends

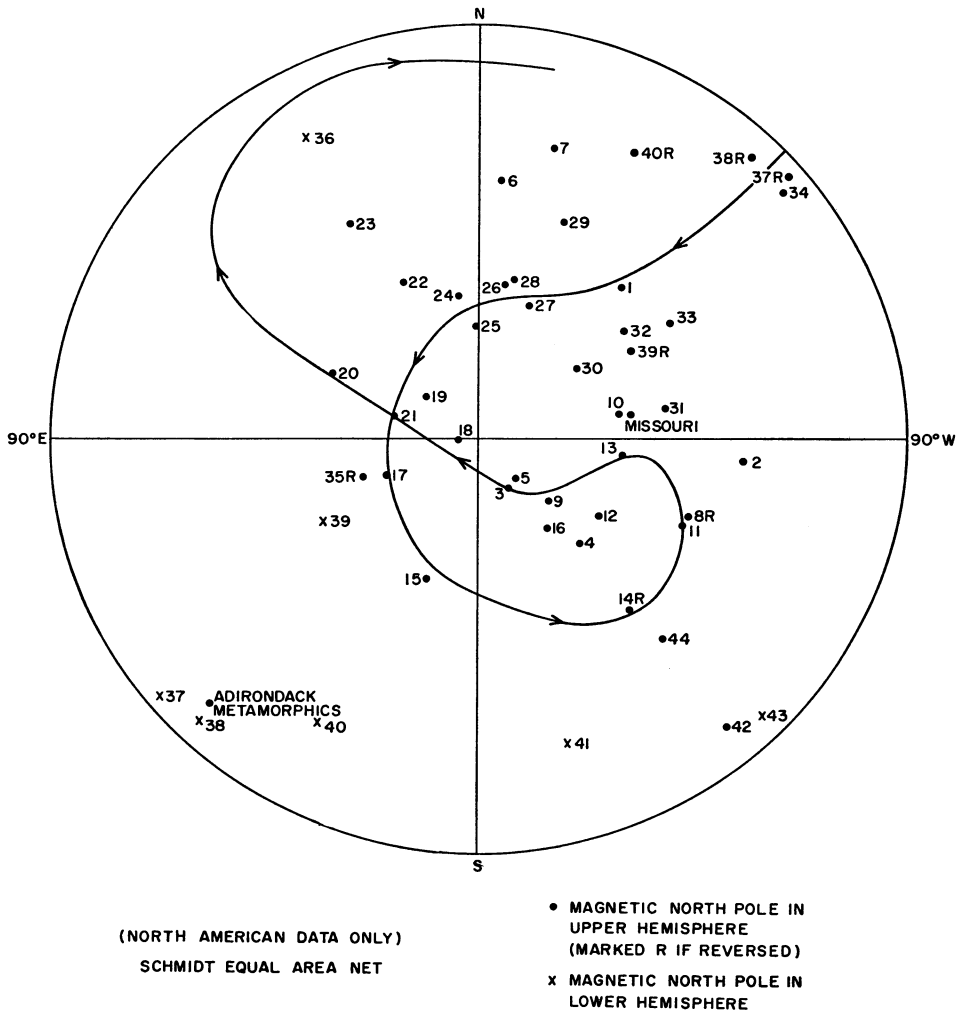


FIG. 7.—Precambrian virtual magnetic poles

This content downloaded from 129.100.058.076 on October 26, 2018 00:31:30 AM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

back to rocks of early Precambrian age. A measurable paleomagnetism appears to be preserved in these rocks, and we conclude that the earth's magnetic field dates back to at least this time in earth history. There seems to be little question that paleomagnetism offers a very useful technique for helping to unravel the complex history of the Precambrian when we have only a few satisfactory techniques.

TABLE 3

POLE POSITIONS OF PRECAMBRIAN ROCKS FROM NORTH AMERICA

No.		Sample No.*	Pole Position	
			Latitude	Longitude
1.....	Hakatai shale	57-D	29½ N.	149 W.
2.....	Hakatai shale and Bass. 1st.	61-D	4 S.	128 W.
3.....	Hakatai shale	63-D	9 S.	174 W.
4.....	Bass limestone	65-D	20 S.	159 W.
5.....	Shinumo quartzite	67-D	7 S.	163
6.....	Hazel	69-D	53 N.	173 W.
7.....	Hazel	71-D	59½ N.	154 E.
8.....	McNamara	72-D R	14 N.	42 E.
9.....	Miller Peak	73-D	11 S.	166 W.
10.....	Spokane sh.	75-D	5 N.	152 W.
11.....	Spokane sh.	76-D	16 S.	139 W.
12.....	Appekunny arg.	77-D	15 S.	156 W.
13.....	Grinell	78-D	3 S.	152 W.
14.....	Bonito Canyon quartzite	79-D R	33 N.	34 E.
15.....	Lodore	23-B	27 S.	169 E.
16.....	Jacobsville (Keweenawan)	22-B	17½ S.	166 W.
17.....	Grenville	21-B	7 S.	162 E.
18.....	Orienta	19-B	0	176 E.
19.....	Freda and Nonesuch	16-B	9 N.	170 E.
20.....	Eileen	17-B	13 N.	150½ E.
21.....	Jacobsville (Sault)	18-B	5 N.	164 E.
22.....	Sault Freda	14-B	31 N.	164½ E.
23.....	Tudor gabbro	43-D	41½ N.	149 E.
24.....	Copper Harbor	13-B	29 N.	176½ E.
25.....	Duluth gabbro	12-B	23½ N.	179½ W.
26.....	Mamainse pt. lavas	11-B	31½ N.	173½ W.
27.....	Portage lake lavas	10-B	26½ N.	168½ W.
28.....	Logan diabase II	9-B	33 N.	171½ W.
29.....	Boulter gabbro	35-D	42½ N.	157 W.
30.....	Normetal E.NE. dike	2-S	13½ N.	159 W.
31.....	Noranda E.-W. dike	5-S	4½ N.	144 W.
32.....	Opawica E.NE. dike	3-S	20½ N.	149 W.
33.....	Noranda E.NE. dike	4-S	20 N.	140 W.
34.....	Sudbury A	52-D	38½ N.	99½ W.
35.....	Thanet and Umfraville	41-D R	8½ N.	22 W.
36.....	Abana N.-S. dike	9-S	57½ N.	56 E.
37.....	Alona Bay lavas	3-B r	39 S.	85 E.
38.....	Michigan dikes	33-D r	45 S.	81 E.
39.....	Sibley	2-B r	16 S.	31 E.
40.....	Logan diabase sills	2-B r	53½ S.	50 E.
41.....	L. Dufault N.-S. dike	11-S	61½ S.	47½ W.
42.....	Aldermac N.-S. dike	12-S	49 S.	105½ W.
43.....	Matachewan dikes (A and B)	13, 14-S	46 S.	81 W.
44.....	Noranda N.-S.	15-S	38 S.	136½ W.
.....	Missouri Precambrian igneous rocks	.....	5 N.	150 W.
.....	Adirondack metamorphics	.....	44 S.	106 E.

* "D" refers to numbers quoted by Cox and Doell (1960); "B" refers to numbers quoted by DuBois (1962); "S" refers to present study. Samples marked "R" have been plotted with a reverse polarity; those marked "r" are shown with a normal and a reverse polarity.

This content downloaded from 129.100.058.076 on October 26, 2018 00:31:30 AM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

CONCLUSIONS

When investigating Precambrian rocks it has been shown that a careful study of remanent magnetism can be a useful method of classification. The direction of remanent magnetism varied widely from dike swarm to dike swarm. In addition, the intensity of remanent magnetism varied from dike swarm to dike swarm in a characteristic fashion. Although it is difficult to draw any real conclusions about the earth's magnetic history, there is little doubt that it existed and that it wandered extensively in Precambrian times.

ACKNOWLEDGMENTS.—Many people contributed to this study. R. M. Farquhar and G. F. West of the University of Toronto provided many helpful discussions. The author received financial assistance from a Garnet W. McKee–Lachlan Gilchrist Scholarship and the National Research Council of Canada. Normetal Mining Corporation and Teck Hughes Gold Mines Ltd. kindly allowed samples to be collected underground. Dr. J. T. Wilson has shown a continuing interest in this study.

REFERENCES CITED

ANDERSON, E. M., 1951, The dynamics of faulting, chap. iii: London, Oliver & Boyd.

BALSLEY, J. R., JAMES, H. L., and WEIR, K. L., 1949, Aeromagnetic survey of parts of Baraga, Iron and Houghton Counties, Michigan, with preliminary geologic interpretation: U.S. Geol. Survey Geophys. Inv.

CAMPBELL, F. A., 1962, Age of mineralization at Quemont and Horne Mines (abs.: Canadian Institute of Mining and Metallurgy Bull., v. 55, p. 627–630.

COX, A., and DOELL, R. R., 1960, Review of paleomagnetism: Geol. Soc. America Bull., v. 71, p. 647–768.

——— — ——— 1961, Paleomagnetism, advances in geophysics: v. 8, New York, Academic Press.

CREER, K., 1962, Jour. Geomag. and Geoel. Kyoto, v. 13, p. 159–165.

DUBOIS, P. M., 1957, Comparison of paleomagnetic results for selected rocks of Great Britain and North America: Advances in Physics, v. 6, p. 177–186.

——— 1962, Paleomagnetism and correlation of Keweenawan rocks: Canada Geol. Survey Bull. 71, 75 pp.

FAIRBAIRN, H. W., HURLEY, P. M., and PINSON, W. H., 1960, Paper presented to Geol. Association of Canada.

GILL, J. E., and L'ESPERANCE, R. L., 1952, Diabase dikes in the Canadian Shield: Royal Soc. Canada Proc., v. 44, ser. 3.

GOLDICH, S. S., NIER, A. O., BAADSGAARD, H., HOFFMAN, J. H., and KRUEGER, H. W., 1961, The Precambrian geology and geochronology of Minnesota: University of Minnesota Bull. 41.

GRAHAM, J. W., 1953, Changes of ferromagnetic minerals and their bearing on magnetic properties of rocks: Jour. Geophys. Research, v. 58, p. 243–260.

HAYS, W., SCHARON, LEROY, and ANDERSON, R. E., 1962, The results of a paleomagnetic investigation of the Precambrian igneous rocks, southeast Missouri: paper presented at the A.G.U.

HOGG, N., 1954, Geology of Godfrey Township: Ontario Dept. of Mines no. 63, pt. 7.

HOOD, P. J., 1958, Paleomagnetic measurements of some Precambrian rocks in Ontario: Ph.D. dissertation, University of Toronto.

——— 1961, Paleomagnetic study of the Sudbury Basin: Jour. Geophys. Research, v. 66, p. 1235–1241.

LOWDON, J. A., 1961, Age determinations by the geological survey of Canada: Canada Geol. Survey Paper 61–17.

MOORE, E. M., 1955, Geology of the Miller Lake portion of the Gowganda silver area: Ontario Dept. of Mines no. 64.

——— and ARMSTRONG, H. S., 1943, Geology of the East Bull Lake area: Ontario Dept. of Mines, v. 52, no. 6.

NEEL, L., 1955, Some theoretical aspects of rock magnetism: Advances in Physics, v. 4.

NORMAN, G. W. H., 1948, Structural geology of some Canadian ore deposits: Canadian Institute of Mining and Metallurgy, Montreal, 1948.

RICHEY, J. E., 1939, The dikes of Scotland: Geol. Soc. Edinburgh Trans., v. 13, p. 393–435.

STRANGWAY, D. W., 1960, Magnetic properties of some Canadian diabase dike swarms: Ph.D. dissertation, University of Toronto.

——— 1961, Magnetic properties of diabase dikes: Jour. Geophys. Research, v. 66, p. 3021–3032.

THELLIER, E., and THELLIER, O., 1959, Sur l'intensité du champ magnétique terrestre dans le passé historique et géologique: Ann. Geophys., v. 15, p. 285–376.

VAN VOORHIS, G., 1964, A method of computing the magnetization for two-dimensional structures: M.Sc. thesis, Michigan Technological University.

WANLESS, R. K., and FAHRIG, W. F. Paper presented at the IUCC meeting, Berkeley, California, 1963.

This content downloaded from 129.100.058.076 on October 26, 2018 00:31:30 AM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).